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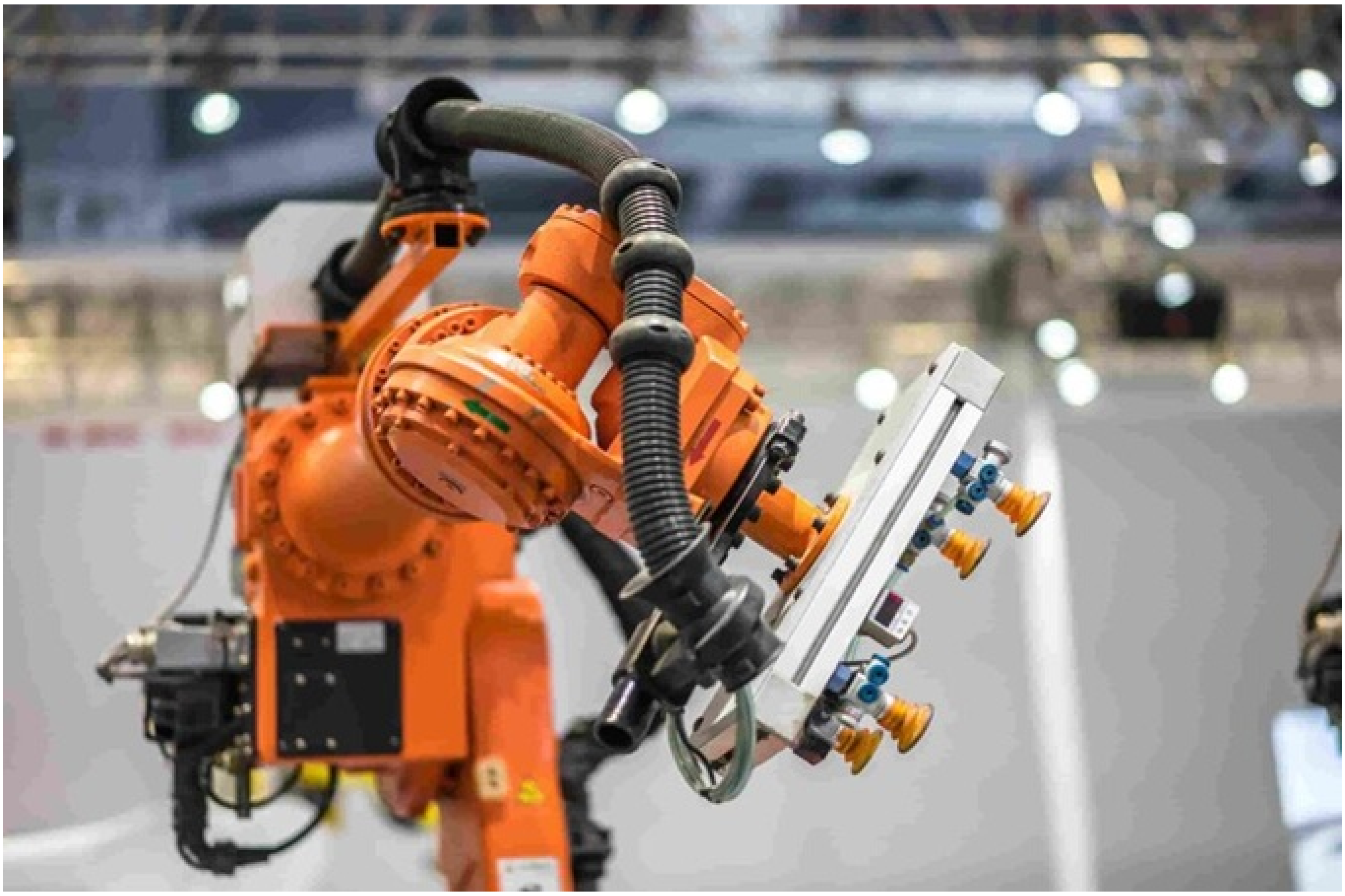
Qualification Pack



Senior Technician - Robotics and Automation

QP Code: TNSD/CSC/Q0401

Version: 1.0

NSQF Level: 4.5

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Contents

TNSD/CSC/Q0401: Senior Technician - Robotics and Automation

Brief Job Description

This qualification comprises various modules essential for the Senior Technician - Mechatronics job role. This qualification is designed to enable learners to achieve the following: Become proficient in automating industrial processes across diverse industry sectors.

Personal Attributes

Secure employment within a broad range of industries, including engineering, manufacturing, process, and automotive sectors.Pursue entrepreneurial opportunities in the field of mechatronics

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. TNSD/CSC/N0409: Assemble and Disassemble Machine Elements.
2. TNSD/CSC/N0410: Design Control Panels.
3. TNSD/CSC/N0411: Program and Troubleshoot Advanced PLC Systems.
4. TNSD/CSC/N0412: Develop Embedded Systems using PIC Microcontrollers.
5. TNSD/CSC/N0413: Apply Industry 4.0 Concepts.
6. TNSD/CSC/N0414: Design and Implement Mechatronics Systems.
7. TNSD/CSC/N0415: Utilize SolidWorks for Electrical and Mechanical Design.
8. TNSD/CSC/N0416: Program and Operate Robotic Systems.
9. TNSD/CSC/N0417: Commission, Troubleshoot, and Maintain Automated Systems
10. TNSD/CSC/N0418: Apply Principles of Management
11. DGT/VSQ/N0101: Employability Skills (30 Hours)
12. TNSD/CSC/N0419: OJT(Industry)

Qualification Pack (QP) Parameters

Sector	Capital Goods

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GOVERNMENT OF INDIA
MINISTRY OF SKILL DEVELOPMENT & ENTREPRENEURSHIP

Qualification Pack

Sub-Sector	
Occupation	Sr. Technician (Robotics and Automation)
Country	India
NSQF Level	4.5
Credits	40
Aligned to NCO/ISCO/ISIC Code	3113.0401
Minimum Educational Qualification & Experience	Previous relevant Qualification of NSQF Level (Electro-Mechanical Technician (level 4))
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	NA
Minimum Job Entry Age	17 Years
Last Reviewed On	NA
Next Review Date	07/10/2028
NSQC Approval Date	07/10/2025
Version	1.0
Reference code on NQR	QG-4.5-CG-04453-2025-V1-TNSDC
NQR Version	1.0

Remarks:

nil

TNSD/CSC/N0409: Assemble and Disassemble Machine Elements.

Description

Able to understand the functioning different machine elements. Able to understand the assembly and disassembly procedures.

Scope

The scope covers the following :

- Understand Basic engineering materials, heat treatment and their applications.
- Understand different Machine Elements Bearings, Axles and Shafts, Couplings, Gears, Screws & Nuts, Belt & Chain, Seals, Lubricants.
- Understand the Terms used in Assembly & Disassembly process.
- Understand the Recycling and disposal of different types of Equipment & Products case studies

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

PC1. Understand the Basics of engineering materials and their applications.
PC2. Understand the Basics of limits , Fits , form & positional tolerances
PC3. Understand the Types of joints
PC4. Understand the Machine Elements – Bearings, Axles, Couplings, Gears, Screws, nuts, Belt, Chain
PC5. Understand the Lubricants and seals
PC6. Understand the functions of Assembly and disassembly
PC7. Understand the disassembly, recycling and disposal of different types of Equipment & Products

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Understand the Basics of engineering materials and their applications
KU2. Understand the Basics of limits , Fits , form & positional tolerances

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Understand the functions of Assembly and disassembly

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	100	**-**	**-**	**-**
PC1. Understand the Basics of engineering materials and their applications.	10	-	-	-
PC2. Understand the Basics of limits , Fits , form & positional tolerances	15	-	-	-
PC3. Understand the Types of joints	15	-	-	-
PC4. Understand the Machine Elements – Bearings, Axles, Couplings, Gears, Screws, nuts, Belt, Chain	15	-	-	-
PC5. Understand the Lubricants and seals	15	-	-	-
PC6. Understand the functions of Assembly and disassembly	15	-	-	-
PC7. Understand the disassembly, recycling and disposal of different types of Equipment & Products	15	-	-	-
NOS Total	**100**	**-**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	TNSD/CSC/N0409
NOS Name	Assemble and Disassemble Machine Elements.
Sector	Capital Goods
Sub-Sector	
Occupation	Sr. Technician (Robotics and Automation)
NSQF Level	4.5
Credits	1
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

TNSD/CSC/N0410: Design Control Panels.

Description

Able to understand the control panel & familiarize the procedures. Able to create a control design and the related components.

Scope

The scope covers the following :

- Understand the designing of Control Panel using Solid works software
- Understand the Control Panel construction using appropriate components
- Understand the Control diagram& Schematic diagram
- Understand the usage of Cabinet layout diagram
- Understand the need of various protection devices
- Understand preparation BOM and reports of the corresponding electrical diagram

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

PC1. Familiarize control panel wiring tools and utilisation
PC2. Understand a Control Panel using appropriate components
PC3. Understand the Designing methods of Control diagram & Schematic diagram
PC4. Understand the Cabinet layout construction
PC5. Understand the need of various protection devices
PC6. Understand the preparation of BOM and reports to the corresponding electrical diagram

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Understand a Control Panel using appropriate components
KU2. Understand the Designing methods of Control diagram & Schematic diagram

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Familiarize control panel wiring tools and utilization
GS2. Understand the Cabinet layout construction

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	50	**40**	**-**	**10**
PC1. Familiarize control panel wiring tools and utilisation	8	5	-	-
PC2. Understand a Control Panel using appropriate components	8	7	-	2
PC3. Understand the Designing methods of Control diagram & Schematic diagram	10	7	-	2
PC4. Understand the Cabinet layout construction	7	7	-	2
PC5. Understand the need of various protection devices	10	7	-	2
PC6. Understand the preparation of BOM and reports to the corresponding electrical diagram	7	7	-	2
NOS Total	**50**	**40**	**-**	**10**

National Occupational Standards (NOS) Parameters

NOS Code	TNSD/CSC/N0410
NOS Name	Design Control Panels.
Sector	Capital Goods
Sub-Sector	
Occupation	Sr. Technician (Robotics and Automation)
NSQF Level	4.5
Credits	3
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

TNSD/CSC/N0411: Program and Troubleshoot Advanced PLC Systems.

Description

Able to understand the different PLC programming methods and troubleshooting a PLC Able to understand the functioning of HMI, Stepper drive and Servo drive.

Scope

The scope covers the following :

- Familiarize the various PLC types and programming methods/languages.
- Understand the communication protocols and trouble shooting.
- Understand the HMI programming
- Understand stepper drives and servo drives interfacing.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

PC1. Understand the various PLC types and programming methods/languages
PC2. Understand the Communication protocols and trouble shoot the same for PLC
PC3. Understand and perform the HMI programming
PC4. Understand and perform the Interfacing of stepper and servo drives
PC5. Understand and perform the programming using analog signals from field devices

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Understand the various PLC types and programming methods/languages
KU2. Understand the Communication protocols and trouble shoot the same for PLC

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Understand and perform the programming using analog signals from field devices

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	50	**40**	**-**	**10**
PC1. Understand the various PLC types and programming methods/languages	10	-	-	-
PC2. Understand the Communication protocols and trouble shoot the same for PLC	10	-	-	-
PC3. Understand and perform the HMI programming	10	10	-	4
PC4. Understand and perform the Interfacing of stepper and servo drives	10	20	-	3
PC5. Understand and perform the programming using analog signals from field devices	10	10	-	3
NOS Total	**50**	**40**	**-**	**10**

National Occupational Standards (NOS) Parameters

NOS Code	TNSD/CSC/N0411
NOS Name	Program and Troubleshoot Advanced PLC Systems.
Sector	Capital Goods
Sub-Sector	
Occupation	Sr. Technician (Robotics and Automation)
NSQF Level	4.5
Credits	5
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

TNSD/CSC/N0412: Develop Embedded Systems using PIC Microcontrollers.

Description

Able to understand the architecture of a PIC microcontroller, Able to perform programming in embedded C Able to understand the communication methods with peripherals.

Scope

The scope covers the following :

- Understand the architecture of microcontrollers
- Understand the programming of PIC microcontroller
- Understand the pin configuration of PIC
- Understanding the interfacing techniques of sensors & peripherals

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

PC1. Understand the concepts of embedded system design
PC2. Learn the architecture of microcontroller
PC3. understand Pin configuration & programming of PIC microcontroller
PC4. Understand Interfacing techniques of sensors & peripherals
PC5. Create a Program by Embedded ‘C’

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Understand the concepts of embedded system design
KU2. Learn the architecture of microcontroller

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Understand Interfacing techniques of sensors & peripherals

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	50	**40**	**-**	**10**
PC1. Understand the concepts of embedded system design	10	-	-	-
PC2. Learn the architecture of microcontroller	10	-	-	-
PC3. understand Pin configuration & programming of PIC microcontroller	10	-	-	-
PC4. Understand Interfacing techniques of sensors & peripherals	10	-	-	-
PC5. Create a Program by Embedded ‘C’	10	40	-	10
NOS Total	**50**	**40**	**-**	**10**

National Occupational Standards (NOS) Parameters

NOS Code	TNSD/CSC/N0412
NOS Name	Develop Embedded Systems using PIC Microcontrollers.
Sector	Capital Goods
Sub-Sector	
Occupation	Sr. Technician (Robotics and Automation), Sr. Technician (Robotics and Automation)
NSQF Level	4.5
Credits	4
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

TNSD/CSC/N0413: Apply Industry 4.0 Concepts.

Description

Able to understand and familiarize the protocols and procedures followed to make smart factories.

Scope

The scope covers the following :

- Understand the smartness in Smart Factories, Smart cities, smart products and smart services
- Understand the various systems used in a manufacturing plant and their role in an Industry 4.0
- Understand the power of Cloud Computing in a networked economy for getting the benefits
- Understand the opportunities, challenges by Industry 4.0

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

PC1. Understand the Basic concept of Industry 4.0

PC2. Understand the Smartness in Smart Factories, Smart cities, smart products and smart services

PC3. : Understand and create the Various systems used in a manufacturing plant and their role in an Industry 4.0 world

PC4. Understand the Power of Cloud Computing in a networked economy and perform programming to synchronize data from cloud network.

PC5. Analyze Opportunities, challenges brought about by Industry 4.0-Applications and perform troubleshooting.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Understand the Basic concept of Industry 4.0

KU2. Understand and create the Various systems used in a manufacturing plant and their role in an Industry 4.0 world

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Analyze Opportunities, challenges brought about by Industry 4.0-Applications and perform troubleshooting.

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	50	**40**	**-**	**10**
PC1. Understand the Basic concept of Industry 4.0	10	-	-	-
PC2. Understand the Smartness in Smart Factories, Smart cities, smart products and smart services	10	-	-	-
PC3. : Understand and create the Various systems used in a manufacturing plant and their role in an Industry 4.0 world	10	10	-	3
PC4. Understand the Power of Cloud Computing in a networked economy and perform programming to synchronize data from cloud network.	10	15	-	3
PC5. Analyze Opportunities, challenges brought about by Industry 4.0-Applications and perform troubleshooting.	10	15	-	4
NOS Total	**50**	**40**	**-**	**10**

National Occupational Standards (NOS) Parameters

NOS Code	TNSD/CSC/N0413
NOS Name	Apply Industry 4.0 Concepts.
Sector	Capital Goods
Sub-Sector	
Occupation	Sr. Technician (Robotics and Automation)
NSQF Level	4.5
Credits	2
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

TNSD/CSC/N0414: Design and Implement Mechatronics Systems.

Description

Able to understand an industry and the application of the automation skills to ensure the factory is completely automated.

Scope

The scope covers the following :

- Understand the key elements for designing the Mechatronics system
- Understand and compare the drive systems
- Understand the design, fabrication and programming of different automation stations
- Understand the operation of the stations in manual mode and auto mode.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

PC1. Understand the Mechatronics Subsystem
PC2. Understand a Control Panel & Create program to interface with PLC
PC3. Understand and Create programs using I/O address, Sensors/Actuator box & Axis Module.
PC4. Understand and Perform programming & operations using Mechatronics system
PC5. Understand the Mechatronics system Sequence &perform Programming.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Understand the Mechatronics Subsystem
KU2. Understand a Control Panel & Create program to interface with PLC

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Understand and Perform programming & operations using Mechatronics system

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	50	**40**	**-**	**10**
PC1. Understand the Mechatronics Subsystem	10	-	-	-
PC2. Understand a Control Panel & Create program to interface with PLC	10	10	-	-
PC3. Understand and Create programs using I/O address, Sensors/Actuator box & Axis Module.	10	10	-	-
PC4. Understand and Perform programming & operations using Mechatronics system	10	10	-	5
PC5. Understand the Mechatronics system Sequence &perform Programming.	10	10	-	5
NOS Total	**50**	**40**	**-**	**10**

National Occupational Standards (NOS) Parameters

NOS Code	TNSD/CSC/N0414
NOS Name	Design and Implement Mechatronics Systems.
Sector	Capital Goods
Sub-Sector	
Occupation	Sr. Technician (Robotics and Automation)
NSQF Level	4.5
Credits	5
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

TNSD/CSC/N0415: Utilize SolidWorks for Electrical and Mechanical Design.

Description

Able to understand and familiarize the Solid works software to design mechanical parts and electrical panels.

Scope

The scope covers the following :

- Familiarize the screen layout and main menu
- Familiarize the tools for sketching
- Familiarize the Modeling
- Understand the concept of assembling & detailed drawing
- Familiarize tools for surface Modeling

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

PC1. Understand screen layout and main menu, and practice tools for sketching, Modelling
PC2. Learn the Safety and management concepts of designing a project
PC3. Familiarize Marking Procedure and perform mounting of electrical devices on the control panel
PC4. Perform the wiring of the field devices with the control panel (control Box)
PC5. Understanding the concept of assembling & detailed drawing
PC6. Familiarization of tools for surface Modelling

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Understand screen layout and main menu, and practice tools for sketching, Modelling
KU2. Learn the Safety and management concepts of designing a project

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Familiarize Marking Procedure and perform mounting of electrical devices on the control panel

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	-	**80**	**-**	**20**
PC1. Understand screen layout and main menu, and practice tools for sketching, Modelling	-	10	-	5
PC2. Learn the Safety and management concepts of designing a project	-	10	-	5
PC3. Familiarize Marking Procedure and perform mounting of electrical devices on the control panel	-	15	-	2
PC4. Perform the wiring of the field devices with the control panel (control Box)	-	15	-	3
PC5. Understanding the concept of assembling & detailed drawing	-	15	-	2
PC6. Familiarization of tools for surface Modelling	-	15	-	3
NOS Total	**-**	**80**	**-**	**20**

National Occupational Standards (NOS) Parameters

NOS Code	TNSD/CSC/N0415
NOS Name	Utilize SolidWorks for Electrical and Mechanical Design.
Sector	Capital Goods
Sub-Sector	
Occupation	Sr. Technician (Robotics and Automation)
NSQF Level	4.5
Credits	3
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

TNSD/CSC/N0416: Program and Operate Robotic Systems.

Description

Able to understand the concepts of robot technology, robot anatomy, programming methods and working of a robot. Able to create and execute programs for robot in teach box and software.

Scope

The scope covers the following :

- Understand the robot anatomy and Kinematics
- Understand the purpose of Drive Units.
- Understand the Controlling of Robots using Open Loop and Closed Loop
- Familiarize the Robot programming
- Understand the function of Robot End Effectors

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

PC1. Understand the technological advance and importance of Robot and kinematics
PC2. Function of different drive and control systems used in Robot application
PC3. Identify the components of Robot
PC4. Understand the purpose of Drive Units.
PC5. Controlling of Robots using Open Loop and Closed Loop
PC6. Familiarizing the programming of Robot
PC7. Understanding the function of Robot End Effectors
PC8. Understand the advantages of Robot in Manufacturing
PC9. Perform programming a robot for given application using teach box and COSIMIR software.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Understand the technological advance and importance of Robot and kinematics
KU2. Understand the purpose of Drive Units.

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Function of different drive and control systems used in Robot application
GS2. Familiarizing the programming of Robot

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	50	**40**	**-**	**10**
PC1. Understand the technological advance and importance of Robot and kinematics	5	-	-	-
PC2. Function of different drive and control systems used in Robot application	5	-	-	-
PC3. Identify the components of Robot	5	-	-	-
PC4. Understand the purpose of Drive Units.	5	-	-	-
PC5. Controlling of Robots using Open Loop and Closed Loop	5	-	-	-
PC6. Familiarizing the programming of Robot	5	-	-	-
PC7. Understanding the function of Robot End Effectors	5	-	-	-
PC8. Understand the advantages of Robot in Manufacturing	5	-	-	-
PC9. Perform programming a robot for given application using teach box and COSIMIR software.	10	40	-	10
NOS Total	**50**	**40**	**-**	**10**

National Occupational Standards (NOS) Parameters

NOS Code	TNSD/CSC/N0416
NOS Name	Program and Operate Robotic Systems.
Sector	Capital Goods
Sub-Sector	
Occupation	Sr. Technician (Robotics and Automation)
NSQF Level	4.5
Credits	4
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

TNSD/CSC/N0417: Commission, Troubleshoot, and Maintain Automated Systems

Description

Able to learn, understand and familiarize the commissioning, troubleshooting and maintenance procedures followed in industries.

Scope

The scope covers the following :

- Understand the overall function and sub-function of a system including its protective equipment.
- Understand information from technical documentation.
- Understand the influence of components on the whole system.
- Understand the Mechatronics systems commissioning and stipulate the approach to be adopted.
- Understand about diagnostic systems, interpret function and error protocols.
- Familiarize the effectiveness of protective measures.
- Understand and use maintenance plans, procedures to ascertain maintenance requirements.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

PC1. Understand the Overall function and sub-function of a system including its protective equipment
PC2. Understand the information from technical documentation for this purpose
PC3. Understand the Maintenance plans, procedures to ascertain maintenance requirements
PC4. Understand the commissioning of a whole Mechatronics system
PC5. Understand the Diagnostic systems, interpret function and error protocols
PC6. Familiarize the effectiveness of protective measures
PC7. Understand the trouble shooting methods

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Understand the Overall function and sub-function of a system including its protective equipment
KU2. Understand the information from technical documentation for this purpose
KU3. Understand the Maintenance plans, procedures to ascertain maintenance requirements

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Familiarize the effectiveness of protective measures

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	100	**-**	**-**	**-**
PC1. Understand the Overall function and sub-function of a system including its protective equipment	10	-	-	-
PC2. Understand the information from technical documentation for this purpose	15	-	-	-
PC3. Understand the Maintenance plans, procedures to ascertain maintenance requirements	15	-	-	-
PC4. Understand the commissioning of a whole Mechatronics system	15	-	-	-
PC5. Understand the Diagnostic systems, interpret function and error protocols	15	-	-	-
PC6. Familiarize the effectiveness of protective measures	15	-	-	-
PC7. Understand the trouble shooting methods	15	-	-	-
NOS Total	**100**	**-**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	TNSD/CSC/N0417
NOS Name	Commission, Troubleshoot, and Maintain Automated Systems
Sector	Capital Goods
Sub-Sector	
Occupation	Sr. Technician (Robotics and Automation)
NSQF Level	4.5
Credits	1
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

TNSD/CSC/N0418: Apply Principles of Management

Description

Understand the management principles, hierarchy, workflow and financials.

Scope

The scope covers the following :

- To understand the meaning and definition of Management
- To understand the meaning and definition of Organization
- Imparting knowledge on planning
- Understanding the meaning and features of Co-ordination & Controlling
- To understand the meaning and definition of staffing
- To understand the meaning, definition and features of Banking
- Knowledge on Company and Trade

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

PC1. Definition of Management & Organization
PC2. Imparting knowledge on planning
PC3. Features of Co-ordination & Controlling
PC4. Meaning and definition of staffing & Banking
PC5. Imparting knowledge on Company and Trade

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Definition of Management and Organization
KU2. Features of Co-ordination & Controlling

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Imparting knowledge on Company and Trade

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	100	**-**	**-**	**-**
PC1. Definition of Management & Organization	20	-	-	-
PC2. Imparting knowledge on planning	20	-	-	-
PC3. Features of Co-ordination & Controlling	20	-	-	-
PC4. Meaning and definition of staffing & Banking	20	-	-	-
PC5. Imparting knowledge on Company and Trade	20	-	-	-
NOS Total	**100**	**-**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	TNSD/CSC/N0418
NOS Name	Apply Principles of Management
Sector	Capital Goods
Sub-Sector	
Occupation	Sr. Technician (Robotics and Automation)
NSQF Level	4.5
Credits	1
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

DGT/VSQ/N0101: Employability Skills (30 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

PC1. understand the significance of employability skills in meeting the job requirements

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.

Basic English Skills

To be competent, the user/individual on the job must be able to:

PC4. speak with others using some basic English phrases or sentences

Communication Skills

To be competent, the user/individual on the job must be able to:

PC5. follow good manners while communicating with others

PC6. work with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

PC7. communicate and behave appropriately with all genders and PwD

PC8. report any issues related to sexual harassment

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

PC9. use various financial products and services safely and securely

PC10. calculate income, expenses, savings etc.

PC11. approach the concerned authorities for any exploitation as per legal rights and laws

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

PC12. operate digital devices and use its features and applications securely and safely

PC13. use internet and social media platforms securely and safely

Entrepreneurship

To be competent, the user/individual on the job must be able to:

PC14. identify and assess opportunities for potential business

PC15. identify sources for arranging money and associated financial and legal challenges

Customer Service

To be competent, the user/individual on the job must be able to:

PC16. identify different types of customers

PC17. identify customer needs and address them appropriately

PC18. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC19. create a basic biodata

PC20. search for suitable jobs and apply

PC21. identify and register apprenticeship opportunities as per requirement

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use basic spoken English language

KU6. Do and dont of effective communication

KU7. inclusivity and its importance

KU8. different types of disabilities and appropriate communication and behaviour towards PwD

KU9. different types of financial products and services

KU10. how to compute income and expenses
KU11. importance of maintaining safety and security in financial transactions
KU12. different legal rights and laws
KU13. how to operate digital devices and applications safely and securely
KU14. ways to identify business opportunities
KU15. types of customers and their needs
KU16. how to apply for a job and prepare for an interview
KU17. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. communicate effectively using appropriate language
GS2. behave politely and appropriately with all
GS3. perform basic calculations
GS4. solve problems effectively
GS5. be careful and attentive at work
GS6. use time effectively
GS7. maintain hygiene and sanitisation to avoid infection

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Introduction to Employability Skills	**1**	**1**	**-**	**-**
PC1. understand the significance of employability skills in meeting the job requirements	-	-	-	-
Constitutional values – Citizenship	**1**	**1**	**-**	**-**
PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices	-	-	-	-
Becoming a Professional in the 21st Century	**1**	**3**	**-**	**-**
PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.	-	-	-	-
Basic English Skills	**2**	**3**	**-**	**-**
PC4. speak with others using some basic English phrases or sentences	-	-	-	-
Communication Skills	**1**	**1**	**-**	**-**
PC5. follow good manners while communicating with others	-	-	-	-
PC6. work with others in a team	-	-	-	-
Diversity & Inclusion	**1**	**1**	**-**	**-**
PC7. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC8. report any issues related to sexual harassment	-	-	-	-
Financial and Legal Literacy	**3**	**4**	**-**	**-**
PC9. use various financial products and services safely and securely	-	-	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. calculate income, expenses, savings etc.	-	-	-	-
PC11. approach the concerned authorities for any exploitation as per legal rights and laws	-	-	-	-
Essential Digital Skills	**4**	**6**	**-**	**-**
PC12. operate digital devices and use its features and applications securely and safely	-	-	-	-
PC13. use internet and social media platforms securely and safely	-	-	-	-
Entrepreneurship	**3**	**5**	**-**	**-**
PC14. identify and assess opportunities for potential business	-	-	-	-
PC15. identify sources for arranging money and associated financial and legal challenges	-	-	-	-
Customer Service	**2**	**2**	**-**	**-**
PC16. identify different types of customers	-	-	-	-
PC17. identify customer needs and address them appropriately	-	-	-	-
PC18. follow appropriate hygiene and grooming standards	-	-	-	-
Getting ready for apprenticeship & Jobs	**1**	**3**	**-**	**-**
PC19. create a basic biodata	-	-	-	-
PC20. search for suitable jobs and apply	-	-	-	-
PC21. identify and register apprenticeship opportunities as per requirement	-	-	-	-
NOS Total	**20**	**30**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0101
NOS Name	Employability Skills (30 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	2
Credits	1
Version	1.0
Last Reviewed Date	12/03/2026
Next Review Date	12/03/2029
NSQC Clearance Date	12/03/2026

TNSD/CSC/N0419: OJT(Industry)

Description

On the job training at industry for understanding the systems and procedures followed.

Scope

The scope covers the following :

- On the job training at industry for understanding the systems and procedures followed.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

PC1. Perform a project using various programs in automation industry to make the system free from errors and self-sustainable.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Perform a project using various programs in automation industry to make the system free from errors and self-sustainable.

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Perform a project using various programs in automation industry to make the system free from errors and self-sustainable.

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	-	**80**	**-**	**20**
PC1. Perform a project using various programs in automation industry to make the system free from errors and self-sustainable.	-	80	-	20
NOS Total	**-**	**80**	**-**	**20**

National Occupational Standards (NOS) Parameters

NOS Code	TNSD/CSC/N0419
NOS Name	OJT(Industry)
Sector	Capital Goods
Sub-Sector	
Occupation	Sr. Technician (Robotics and Automation)
NSQF Level	4.5
Credits	10
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

<1. Assessment System Overview:

• Batches assigned to the assessment agencies for conducting the assessment on SIP or email

• Assessment agencies send the assessment confirmation to VTP/TC looping TNSDC

• Assessment agency deploys the ToA certified Assessor for executing the assessment

• TNSDC monitors the assessment process & records

2. Testing Environment:

• Check the Assessment location, date and time

• If the batch size is more than 30, then there should be 2 Assessors.

• Check that the allotted time to the candidates to complete Theory & Practical Assessment is correct.

3. Assessment Quality Assurance levels/Framework:

• Question bank is created by the Subject Matter Experts (SME) are verified by the other SME

• Questions are mapped to the specified assessment criteria

• Assessor must be ToA certified & trainer must be ToT Certified

4. Types of evidence or evidence-gathering protocol:

• Time-stamped & geotagged reporting of the assessor from assessment location

• Centre photographs with signboards and scheme specific branding

5. Method of verification or validation:

• Surprise visit to the assessment location

6. Method for assessment documentation, archiving, and access

• Hard copies of the documents are stored

On the Job:

1. Each module will be assessed separately.

2. The candidate must score 60% in each module to successfully complete the OJT.

3. Tools of Assessment that will be used for assessing whether the candidate is having desired skills and etiquette of dealing with customers, understanding needs & requirements, assessing the customer and perform Soft Skills effectively:

• Videos of Trainees during OJT

4. Assessment of each Module will ensure that the candidate is able to:

• Effective engagement with the customers

• Understand the working of various tools and equipment

Minimum Aggregate Passing % at QP Level : 60

(**Please note**: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
TNSD/CSC/N0409.Assemble and Disassemble Machine Elements.	100	-	-	-	100	5
TNSD/CSC/N0410.Design Control Panels.	50	40	-	10	100	10
TNSD/CSC/N0411.Program and Troubleshoot Advanced PLC Systems.	50	40	-	10	100	10
TNSD/CSC/N0412.Develop Embedded Systems using PIC Microcontrollers.	50	40	-	10	100	10
TNSD/CSC/N0413.Apply Industry 4.0 Concepts.	50	40	-	10	100	10
TNSD/CSC/N0414.Design and Implement Mechatronics Systems.	50	40	-	10	100	10
TNSD/CSC/N0415.Utilize SolidWorks for Electrical and Mechanical Design.	-	80	-	20	100	10
TNSD/CSC/N0416.Program and Operate Robotic Systems.	50	40	-	10	100	10
TNSD/CSC/N0417.Commission, Troubleshoot, and Maintain Automated Systems	100	-	-	-	100	5
TNSD/CSC/N0418.Apply Principles of Management	100	-	-	-	100	5
DGT/VSQ/N0101.Employability Skills (30 Hours)	20	30	-	-	50	5
TNSD/CSC/N0419.OJT(Industry)	-	80	-	20	100	10
Total	**620**	**430**	**-**	**100**	**1150**	**100**

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.